

Arrays, Rows, and Columns Mission

Warm-up and worked example

Name: _____

Date: _____

PAGE 1 | WEEK 3, DAY 2

YOUR MISSION Represent multiplication with arrays and connect turn-around facts.

SMART MOVE Draw groups or an array before relying on memory.

Math Talk

How are 3×8 and 8×3 alike and different?

My idea:

New Words

equal groups	factor	product
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Build a 3-row, 8-column array.
2. Rotate it to show 8 rows of 3.

What the example shows: _____

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Learn it and show your thinking

Name: _____

Date: _____

PAGE 2 | WEEK 3, DAY 2

YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Build a 3-row, 8-column array.

[] **Step 2:** Rotate it to show 8 rows of 3.

[] **Step 3:** Discuss why the total stays 24.

[] **Step 4:** Connect arrays to rectangles and future area work.

[] **Step 5:** Name the commutative property informally as a turn-around fact.

My model, drawing, or organized work:

The big idea in my own words:

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Practice lab

Name: _____

Date: _____

PAGE 3 | WEEK 3, DAY 2

YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. Draw arrays for 4×6 , 2×9 , and 5×7 .

2. Write two equations for each array.

3. Find missing dimensions for arrays with 18 and 20 tiles.

Choose one answer and prove it another way:

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Challenge and final check

Name: _____

Date: _____

PAGE 4 | WEEK 3, DAY 2

YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

How many different rectangular arrays can be made with 24 tiles? Rotations count as the same rectangle.

My plan:

My solution and proof:

Final Check

Draw one array that proves $7 \times 4 = 4 \times 7$.

☐ I can teach it

☐ I need practice

☐ I need help

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Reusable math tool

Name: _____

Date: _____

PAGE 5 | WEEK 3, DAY 2

YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Draw groups or an array before relying on memory.

Grid Lab

What does your drawing prove?
